(U) Verification and Validation Plan

(U) T-150/TRV-150 Heavy-Lift UAS
AFSIM Simulation Model v2.0
(U) AFSIM Project 005



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Version	Date	Author	Description of Change
1.0	[PLACEHOLDER][PLACEHOLDER]		Initial V&V Plan release per MIL-STD-3022 Appendix B. Covers all five plugins and four test families.
			UNCLASSIFIED

(U) Table of Contents

1 (U) Executive Summary 1

2 (U) Problem Statement 2

2.1 (U) Intended Use . . . 2

2.2 (U) M&S Overview . . . 2

2.3 (U) M&S Application . . . 3

2.4 (U) Accreditation Scope . . . 3

2.5 (U) V&V Scope . . . 3

3 (U) M&S Requirements and Acceptability Criteria 4

4 (U) M&S Assumptions, Capabilities, Limitations, and Risks/Impacts 6

4.1 (U) Assumptions . . . 6

4.2 (U) Capabilities . . . 6

4.3 (U) Limitations . . . 7

4.4 (U) Risks/Impacts . . . 7

5 (U) V&V Methodology 8

5.1 (U) V&V Approach . . . 8

5.2 (U) Verification Tasks . . . 8

5.2.1 (U) Code Review . . . 8

5.2.2 (U) Unit-Level Testing . . . 9

5.2.3 (U) Integration Testing . . . 9

5.2.4 (U) Numerical Stability Testing . . . 9

5.3 (U) Validation Tasks . . . 9

5.3.1 (U) Manufacturer Specification Comparison . . . 9

5.3.2 (U) Analytical Prediction Comparison . . . 10

5.3.3 (U) Energy Conservation Validation . . . 10

5.3.4 (U) Pattern Efficiency Analysis . . . 10

5.4 (U) Test Families . . . 11

5.4.1 (U) TF01: 6DOF Physics (5 Tests) . . . 11

5.4.2 (U) TF02: Battery/Power System (5 Tests) . . . 11
5.4.3 (U) TF03: VRS Detection (3 Tests) . . . 12
5.4.4 (U) TF04: Slung Load Dynamics (3 Tests) . . . 12

6 (U) V&V Issues **13**

6.1 (U) Absence of Flight Test Data . . . 13
6.2 (U) Flight Endurance Gap . . . 13
6.3 (U) Estimated Parameters . . . 13
6.4 (U) No Independent V&V Agent . . . 13

7 (U) Key Participants **14**

8 (U) Planned V&V Resources **15**

8.1 (U) Schedule . . . 15
8.2 (U) Computing Resources . . . 15
8.3 (U) Documentation Deliverables . . . 16
8.4 (U) Level of Effort Estimate . . . 16

A (U) M&S Description **17**

B (U) M&S Requirements Traceability Matrix **18**

C (U) Basis of Comparison **19**

C.1 (U) Manufacturer Specifications . . . 19
C.2 (U) Analytical Predictions . . . 19
C.3 (U) Published Literature . . . 20

D (U) References **21**

E (U) Acronyms **22**

F (U) Acronyms **24**

(U) List of Tables

1 (U) Document Information i
2 (U) Plugin Suite Overview 2
3 (U) M&S Requirements 4
4 (U) Model Assumptions 6
5 (U) Model Limitations 7
6 (U) V&V Risks 7
7 (U) Code Review Scope 8
8 (U) Manufacturer Specification Comparison 9
9 (U) Analytical Predictions 10
10 (U) TF01: 6DOF Physics Tests 11
11 (U) TF02: Battery/Power System Tests 11
12 (U) TF03: VRS Detection Tests 12
13 (U) TF04: Slung Load Dynamics Tests 12
14 (U) Key Participants 14
15 (U) V&V Schedule 15
16 (U) Computing Resources 15
17 (U) Documentation Deliverables 16
18 (U) Level of Effort 16
19 (U) Requirements Traceability Matrix 18
20 (U) Manufacturer Specifications 19
21 (U) Analytical Predictions for Validation 19
22 (U) Literature References 20
23 (U) Acronyms 22

1 (U) Executive Summary

(U) This Verification and Validation (V&V) Plan establishes the framework, requirements, methodology, and acceptance criteria for the formal V&V of the T-150/TRV-150 Heavy-Lift UAS AFSIM Simulation Model, version 2.0. The plan is prepared in accordance with MIL-STD-3022, "Documentation of Verification, Validation, and Accreditation (VV&A) for Models and Simulations" [2], and supports the accreditation process described in DoDI 5000.61, "DoD Modeling and Simulation Verification, Validation, and Accreditation" [1].

(U) The T-150 UAS AFSIM Plugin Suite v2.0 is a high-fidelity simulation of the T-150/TRV-150 heavy-lift coaxial octocopter, implemented as five custom C++ plugins within the AFSIM 2.2505 framework. The model encompasses 7,736 lines of code distributed across 26 source files, organized into five functional plugins: a physics-based 6DOF mover, a Li-Ion battery model with voltage sag, an electric propulsion system with thermal dynamics, a Vortex Ring State detection and degradation model, and a cable-suspended slung load dynamics model.

(U) The intended use of this M&S is to support intelligence analysis of T-150 UAS capabilities, mission planning assessments, Counter-UAS (C-UAS) evaluation, and vulnerability analysis. The model is developed for the intelligence community to answer questions regarding heavy-lift UAS operational performance, flight endurance, payload capacity, and vulnerability conditions.

(U) This V&V Plan defines thirteen (13) quantitative requirements with measurable acceptability criteria spanning all five plugin subsystems. Requirements address battery energy conservation, voltage sag fidelity, motor thermal stability, 6DOF position hold and velocity tracking, numerical robustness, VRS state transition accuracy, ground effect modeling, flight endurance, slung load pendulum dynamics, attitude recovery, cruise speed, and maximum payload capacity. Each requirement is traceable to a specific validation test or set of tests.

(U) The V&V approach employs a combination of verification techniques (code review, unit testing, integration testing, numerical stability testing) and validation techniques (comparison to manufacturer specifications, comparison to analytical first-principles predictions, energy conservation analysis, and pattern efficiency analysis). Testing is organized into four test families: TF01 (6DOF Physics), TF02 (Battery/Power System), TF03 (VRS Detection), and TF04 (Slung Load Dynamics), comprising a total of sixteen (16) individual tests.

(U) A key challenge for this V&V effort is the absence of T-150 flight test data. All validation comparisons are made against manufacturer specifications from open-source product data, analytical predictions derived from first-principles physics (momentum theory, Li-Ion electrochemistry, pendulum dynamics), and published aerodynamic literature. A known gap exists between simulated flight endurance (14.3 minutes) and the manufacturer-stated specification (36 minutes), which has been documented and bounded but not fully resolved.

2 (U) Problem Statement

2.1 (U) Intended Use

(U) The T-150 UAS AFSIM Plugin Suite v2.0 is intended for the following analytical applications:

1. (U) **Intelligence Analysis**: Assessing the operational capabilities, performance envelope, and limitations of the T-150/TRV-150 heavy-lift UAS to support intelligence community questions regarding UAS threats. This includes evaluating flight dynamics, payload capacity, energy consumption profiles, and degraded-mode behavior under various operational conditions.
2. (U) **Mission Planning**: Evaluating flight endurance, range, energy consumption, and payload capacity under various mission profiles. The simulation enables analysts to construct mission timelines, estimate fuel (battery) consumption for specific flight profiles, and identify operational constraints for route planning.
3. (U) **Counter-UAS (C-UAS) Assessment**: Providing high-fidelity UAS flight dynamics to support evaluation of detection, tracking, and engagement scenarios against heavy-lift UAS threats.
4. (U) **Vulnerability Analysis**: Analyzing system degradation behavior including Vortex Ring State onset and development, battery depletion effects on voltage and available power, motor thermal limits, slung load instability, and payload-induced performance degradation.

2.2 (U) M&S Overview

Table 2: (U) Plugin Suite Overview

UNCLASSIFIED				
Plugin	AFSIM Type Name	Purpose	Files	LOC
rotorcraft_6dof	Rotorcraft6DOF_Mover	Physics-based 6DOF flight dynamics with rigid body, rotor model, and flight controller	8	∼3,200
t150_battery	T150_BATTERY_PROCESSOR	Li-Ion battery with SOC tracking, OCV polynomial, voltage sag, internal resistance	4	∼1,100
t150_propulsion	T150_PROPULSION_PROCESSOR	Motor power, thermal dynamics, RPM, ground effect, payload, dynamic power	4	∼1,500
				UNCLASSIFIED

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Table 2: (U) Plugin Suite Overview (Continued)

UNCLASSIFIED				
Plugin	AFSIM Type Name	Purpose	Files	LOC
t150_vrs	T150_VRS_ PROCESSOR	VRS detection via state machine, margin calculation, lift/control degradation	4	∼900
t150_slung_load	T150_SLUNG_ LOAD_PROCESSOR	Spherical pendulum slung load, cable tension, resonance detection	4	∼1,000
Total			**26**	**∼7,736**
				UNCLASSIFIED

(U) **Framework**: AFSIM 2.2505 (Advanced Framework for Simulation, Integration, and Modeling) [7]

2.3 (U) M&S Application

(U) The T-150 UAS AFSIM Plugin Suite supports intelligence assessments conducted by [PLACEHOLDER - Organization] regarding the operational capabilities of heavy-lift UAS platforms. Specific analytical products supported include operational performance assessments, flight envelope characterization, energy consumption analysis by mission profile, vulnerability assessments, and C-UAS scenario development.

2.4 (U) Accreditation Scope

(U) This V&V Plan supports full model accreditation of the T-150 UAS AFSIM Plugin Suite v2.0 for intelligence analysis use as described in Section 1.1. The accreditation encompasses all five plugins and their integrated behavior within AFSIM 2.2505.

2.5 (U) V&V Scope

(U) The V&V effort encompasses:

- (U) **Verification**: Confirmation that the model correctly implements the intended physics, algorithms, and data flows as specified in the ADD and UAM. This includes code review, unit-level testing, integration testing, and numerical stability testing.
- (U) **Validation**: Determination that the model produces results consistent with the intended use by comparison against manufacturer specifications, analytical predictions from first-principles physics, and published literature.

3 (U) M&S Requirements and Acceptability Criteria

Table 3: (U) M&S Requirements

UNCLASSIFIED				
Req ID	Requirement Description	Acceptability Criterion	Test Family	Priority
R-001	Battery SOC shall track energy with high accuracy	Energy error $< 0.1\%$ over full discharge	TF02	Critical
R-002	Voltage sag shall match Li-Ion characteristics	Terminal voltage follows OCV minus IR drop	TF02	High
R-003	Motor thermal model shall reach stable equilibrium	Temperature converges within 300 s; no runaway	TF02	High
R-004	6DOF mover shall maintain position hold stability	Position drift < 1 m during steady-state hover	TF01	Critical
R-005	6DOF mover shall track commanded velocity	Velocity error < 1 m/s during cruise	TF01	High
R-006	All plugins shall maintain numerical stability	No NaN, no divergence, no crashes under extremes	TF01–TF04	Critical
R-007	VRS detection shall trigger correct state transitions	CAUTION at margin < 0.8; ONSET at margin < 0.5	TF03	High
R-008	Ground effect shall produce thrust augmentation below 1D	GE factor > 1.0 below 0.76 m AGL; max ≈ 1.3	TF01	Medium
R-009	Flight endurance shall fall within calibrated range	Gap documented with identified contributing factors	TF02	High
R-010	Slung load pendulum frequency shall match analytic prediction	Measured frequency within 5% of $f_n = \frac{1}{2\pi}\sqrt{g/L}$	TF04	High
R-011	6DOF mover shall recover from attitude disturbance	Return to within 5° and 2 m within 10 s	TF01	Medium
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Table 3: (U) M&S Requirements (Continued)

UNCLASSIFIED				
Req ID	Requirement Description	Acceptability Criterion	Test Family	Priority
R-012	Cruise speed shall match manufacturer specification	Steady-state cruise within 1 m/s of 30 m/s	TF01	High
R-013	Maximum payload shall be correctly modeled	Model correctly represents 68 kg max payload	TF02	High
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4 (U) M&S Assumptions, Capabilities, Limitations, and Risks/Impacts

4.1 (U) Assumptions

Table 4: (U) Model Assumptions

UNCLASSIFIED		
ID	Assumption	Impact on Results
A-001	**Rigid body dynamics**: No structural flexibility or aeroelastic effects	Eliminates high-frequency structural dynamics; acceptable for mission-level analysis
A-002	**Standard atmosphere**: Air density at standard sea-level conditions ($\rho = 1.225$ kg/m^3)	Overestimates performance at high altitude; acceptable for low-altitude scenarios
A-003	**Symmetric inertia tensor**: Products of inertia are zero	Eliminates cross-coupling terms; valid for symmetric octocopter
A-004	**No wind model**: Platform operates in still air	Optimistic endurance and stability results
A-005	**Perfect sensors**: All measurements noise-free and lag-free	Overestimates control precision
A-006	**Uniform battery cells**: All cells identical in capacity and resistance	Eliminates cell-level failure modes
A-007	**Flat terrain**: No terrain elevation model	Ground effect assumes flat ground
A-008	**Estimated inertia values**: MOI estimated from geometry, not measured	Affects transient dynamic response
		UNCLASSIFIED

4.2 (U) Capabilities

1. (U) **Full 6DOF flight dynamics**: Newton-Euler rigid body physics with quaternion attitude representation, eight individually modeled rotors, PID-based flight controller, and waypoint navigation with spline interpolation.
2. (U) **Battery voltage sag modeling**: SOC-dependent open-circuit voltage via polynomial curve, SOC-dependent internal resistance, quadratic current solution for stable voltage sag calculation.
3. (U) **VRS detection and modeling**: Five-state machine driven by descent rate and forward speed margins, lift degradation factors (down to 0.4x), and control authority degradation factors.
4. (U) **Slung load dynamics**: Two-DOF spherical pendulum model with cable tension calculation, slack and overload detection, resonance detection, and symplectic integration.
5. (U) **Dynamic power modeling**: Flight-state-dependent power calculation incorporating induced power, climb power, parasite drag power, and acceleration power.

6. (U) **Ground effect**: Cheeseman-Bennett approximation with up to 30% thrust increase at ground level.
7. (U) **Motor thermal dynamics**: First-order thermal model with heat generation, forced convection cooling, and over-temperature warning.

4.3 (U) Limitations

Table 5: (U) Model Limitations

UNCLASSIFIED			
ID	Limitation	Risk	Impact
L-001	No wind model	Medium	Endurance and range estimates are optimistic
L-002	VRS is informational only	Medium	VRS factors computed but do not modify flight dynamics
L-003	Slung load forces do not feed back to mover	Medium	Pendulum forces published but do not alter trajectory
L-004	Flight endurance gap (14.3 vs. 36 min)	Medium	Direct time-based analysis requires calibration factor
L-005	Constant air density	Low	Performance does not degrade with altitude
L-006	No structural failure modes	Low	Single-point failures not assessable
			UNCLASSIFIED

4.4 (U) Risks/Impacts

Table 6: (U) V&V Risks

UNCLASSIFIED			
Risk	Probability	Impact	Mitigation
Endurance gap limits time-based analysis	High	Medium	Document calibration factor (≈2.5x); recommend speed-based metrics
No flight test data for direct validation	High	Medium	Multiple independent validation approaches; classify confidence levels
Estimated parameters may not match actual T-150	Medium	Medium	Sensitivity analysis on key parameters; bound uncertainty ranges
Self-testing reduces V&V independence	Medium	Low	Objective quantitative criteria; recommend independent review
			UNCLASSIFIED

5 (U) V&V Methodology

5.1 (U) V&V Approach

(U) **Verification** establishes that the code correctly implements the intended algorithms and physics. Verification activities include:

- (U) Peer code review of algorithms against design documentation
- (U) Unit-level numerical checks of mathematical functions
- (U) Integration testing of inter-plugin data flow via AuxData
- (U) Boundary and stress testing for numerical robustness

(U) **Validation** establishes that the model's outputs are consistent with real-world behavior within acceptable tolerances. Validation employs three complementary approaches:

1. (U) **Comparison to manufacturer specifications**: Cruise speed, maximum payload, battery capacity, and stated flight time [11]
2. (U) **Comparison to analytical predictions**: Hover power from momentum theory, pendulum natural frequency, VRS onset boundaries [9, 10]
3. (U) **Energy conservation and internal consistency**: Verifying that energy in equals energy out and that state transitions occur at predicted thresholds

5.2 (U) Verification Tasks

5.2.1 (U) Code Review

(U) All 26 source files across the five plugins shall be reviewed for correctness of physics equations, proper unit conversions, numerical guard conditions, proper initialization, and correct AFSIM API usage.

Table 7: (U) Code Review Scope

UNCLASSIFIED		
Plugin	Source Files	Key Algorithms to Review
rotorcraft_6dof	RigidBody6DOF, RotorModel, Mover, MathUtils	Quaternion integration, Newton-Euler, rotor thrust/torque, PID controllers
t150_battery	BatteryProcessor	OCV polynomial, quadratic current solver, SOC integration
		UNCLASSIFIED

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Table 7: (U) Code Review Scope (Continued)

UNCLASSIFIED		
Plugin	Source Files	Key Algorithms to Review
t150_propulsion	PropulsionProcessor	Momentum theory thrust, efficiency curves, thermal dynamics, ground effect
t150_vrs	VRSProcessor	VRS margin calculation, state machine transitions
t150_slung_load	SlungLoadProcessor	Pendulum equations of motion, symplectic integration, cable tension
		UNCLASSIFIED

5.2.2 (U) Unit-Level Testing

(U) Unit-level tests verify individual mathematical functions in isolation: OCV polynomial evaluation, quadratic current solver, thrust coefficient calculation, pendulum angular acceleration, VRS margin calculation, and quaternion operations.

5.2.3 (U) Integration Testing

(U) Integration tests verify inter-plugin communication and data consistency: propulsion-to-battery power flow, mover-to-VRS velocity flow, mover-to-slung-load acceleration flow, and battery SOC depletion rate consistency.

5.2.4 (U) Numerical Stability Testing

(U) Stress tests under boundary conditions: zero payload, maximum payload, overload (>68 kg), battery SOC at 0% and 100%, extreme descent rates (>8 m/s), near-zero altitude, and extreme time step sizes.

5.3 (U) Validation Tasks

5.3.1 (U) Manufacturer Specification Comparison

Table 8: (U) Manufacturer Specification Comparison

UNCLASSIFIED		
Parameter	Manufacturer Value	Acceptance Criterion
Cruise speed	30 m/s	Within 1 m/s
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Table 8: (U) Manufacturer Specification Comparison (Continued)

UNCLASSIFIED		
Parameter	Manufacturer Value	Acceptance Criterion
Maximum payload	68 kg (150 lbs)	Exact match (model parameter)
Battery capacity	∼7 kWh	Exact match (model parameter)
Battery configuration	13S Li-Ion	Exact match (model parameter)
Flight time	36 minutes	Document gap; provide calibration
MTOW	123 kg	Within 5 kg
		UNCLASSIFIED

5.3.2 (U) Analytical Prediction Comparison

Table 9: (U) Analytical Predictions

UNCLASSIFIED			
Prediction	Analytical Formula	Expected Value	Acceptance Criterion
Hover power	$P = T \cdot v_i$, where $v_i = \sqrt{T/(2\rho A)}$	∼17–20 kW	Within 15%
Pendulum frequency	$f_n = \frac{1}{2\pi}\sqrt{g/L}$	0.223 Hz (L=5 m)	Within 5%
VRS onset descent rate	v_{desc} approaching v_i	∼4 m/s	Correct detection
Ground effect factor	$1/(1-(R/4h)^2)$	∼1.3 at $h = R/4$	Within 10%
Bank angle thrust loading	$T = mg/\cos\phi$	1.15x at 30°	Within 5%
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5.3.3 (U) Energy Conservation Validation

(U) Energy conservation testing verifies that total energy consumed equals the integral of power over time. Acceptance criterion: energy balance error $<$ 0.1%.

5.3.4 (U) Pattern Efficiency Analysis

(U) Flight pattern efficiency analysis validates that relative energy cost of different maneuver patterns follows physically expected trends: straight-line (baseline) $<$ sawtooth $<$ sine wave $<$ cycloid $<$ square wave.

5.4 (U) Test Families

5.4.1 (U) TF01: 6DOF Physics (5 Tests)

Table 10: (U) TF01: 6DOF Physics Tests

UNCLASSIFIED			
Test ID	Test Name	Req	Description
TF01-01	Hover Stability	R-004	Command hover at fixed position for 60 s. Measure drift after 10 s transient. Verify drift < 1 m.
TF01-02	Cruise Velocity Tracking	R-005, R-012	Command cruise at 30 m/s. Measure achieved speed. Verify error < 1 m/s.
TF01-03	Attitude Recovery	R-011	Apply asymmetric disturbance. Verify return to level hover within 10 s.
TF01-04	Ground Effect Validation	R-008	Hover at 0.2, 0.5, 1.0, 2.0 m AGL. Compare GE factor to Cheeseman-Bennett.
TF01-05	Numerical Robustness	R-006	Execute extreme maneuver sequences. Verify no NaN or divergence.
			UNCLASSIFIED

5.4.2 (U) TF02: Battery/Power System (5 Tests)

Table 11: (U) TF02: Battery/Power System Tests

UNCLASSIFIED			
Test ID	Test Name	Req	Description
TF02-01	Energy Conservation	R-001	Full discharge from 100% to 0% SOC. Verify energy error $< 0.1\%$.
TF02-02	Voltage Sag	R-002	Hover at multiple SOC levels. Verify $V = OCV - IR$ relationship.
TF02-03	Motor Thermal Equilibrium	R-003	Sustained hover for 300 s. Verify temperature convergence.
TF02-04	Flight Endurance	R-009	Hover until depletion. Document endurance and gap analysis.
TF02-05	Payload Capacity	R-013	Configure payload at 0, 34, 68, 75 kg. Verify overload flag at 75 kg.
			UNCLASSIFIED

5.4.3 (U) TF03: VRS Detection (3 Tests)

Table 12: (U) TF03: VRS Detection Tests

UNCLASSIFIED			
Test ID	Test Name	Req	Description
TF03-01	VRS State Transitions	R-007	Controlled descent at increasing rates with zero forward speed. Verify state transitions.
TF03-02	Forward Speed Immunity	R-007	Descent at 5 m/s with forward speeds of 0, 2, 5, 10 m/s. Verify VRS suppression above 5 m/s.
TF03-03	Numerical Robustness (VRS)	R-006	Extreme descent rates and rapid transitions. Verify no stuck states.
			UNCLASSIFIED

5.4.4 (U) TF04: Slung Load Dynamics (3 Tests)

Table 13: (U) TF04: Slung Load Dynamics Tests

UNCLASSIFIED			
Test ID	Test Name	Req	Description
TF04-01	Pendulum Frequency	R-010	5 m cable, 25 kg load. Apply impulse. Verify frequency within 5% of 0.223 Hz.
TF04-02	Load State Transitions	R-010	Verify DETACHED $\rightarrow$ ATTACHED $\rightarrow$ SWINGING $\rightarrow$ RESONANCE transitions.
TF04-03	Numerical Robustness	R-006	Extreme cable lengths (0.5–20 m), masses (1–55 kg), rapid accelerations.
			UNCLASSIFIED

6 (U) V&V Issues

6.1 (U) Absence of Flight Test Data

(U) No T-150 flight test data is available for direct model-versus-test validation. All validation is performed against manufacturer specifications, analytical predictions, and published literature.

(U) **Impact**: Validation confidence is bounded by the quality of analytical predictions and manufacturer specifications.

(U) **Mitigation**: Multiple independent validation approaches are used. Confidence levels are explicitly stated per capability.

6.2 (U) Flight Endurance Gap

(U) Simulated endurance of 14.3 minutes significantly differs from the 36-minute manufacturer specification. Contributing factors include hover power assumptions and absence of manufacturer-specific motor efficiency data.

(U) **Impact**: Direct use of simulated endurance for absolute time-based analysis produces non-conservative results.

(U) **Mitigation**: The gap is documented with a calibration factor. Analysts instructed to use relative metrics.

6.3 (U) Estimated Parameters

(U) Several parameters are estimated from class-level data rather than measured from the actual T-150: moments of inertia, motor thermal mass, battery internal resistance, and drag coefficient.

(U) **Impact**: Transient dynamics may differ from actual T-150 behavior.

(U) **Mitigation**: Sensitivity analysis recommended. Parameter values documented with sources.

6.4 (U) No Independent V&V Agent

(U) V&V is performed by the model developer rather than by an independent agent, introducing potential confirmation bias.

(U) **Impact**: Reduced independence may limit accreditation authority's confidence.

(U) **Mitigation**: All activities use objective, quantitative criteria defined in advance. Results fully documented for independent review.

7 (U) Key Participants

Table 14: (U) Key Participants

UNCLASSIFIED			
Role	Organization	Individual	Responsibility
V&V Agent	[PLACEHOLDER]	[PLACEHOLDER]	Plan, execute, and document V&V activities
M&S Developer	[PLACEHOLDER]	[PLACEHOLDER]	Develop and maintain the plugin suite
Accreditation Authority	[PLACEHOLDER]	[PLACEHOLDER]	Review V&V results and make accreditation determination
M&S Proponent	[PLACEHOLDER]	[PLACEHOLDER]	Define intended use and analytical requirements
SME (Aerodynamics)	[PLACEHOLDER]	[PLACEHOLDER]	Review 6DOF physics, VRS, and ground effect
SME (Power Systems)	[PLACEHOLDER]	[PLACEHOLDER]	Review battery, propulsion, and thermal modeling
SME (UAS Operations)	[PLACEHOLDER]	[PLACEHOLDER]	Review operational fidelity
Configuration Manager	[PLACEHOLDER]	[PLACEHOLDER]	Maintain version control
			UNCLASSIFIED

8 (U) Planned V&V Resources

8.1 (U) Schedule

Table 15: (U) V&V Schedule

UNCLASSIFIED		
Milestone	Planned Date	Description
V&V Plan Approval	[PLACEHOLDER]	Plan reviewed and approved
Code Review Complete	[PLACEHOLDER]	All 26 source files reviewed
Unit Testing Complete	[PLACEHOLDER]	Unit-level verification tests
Integration Testing Complete	[PLACEHOLDER]	Inter-plugin communication tests
Validation Testing Complete	[PLACEHOLDER]	All 16 tests across 4 families
V&V Report Draft	[PLACEHOLDER]	Draft submitted for review
V&V Report Final	[PLACEHOLDER]	Final delivered to Accreditation Authority
Accreditation Decision	[PLACEHOLDER]	Accreditation determination rendered
		UNCLASSIFIED

8.2 (U) Computing Resources

Table 16: (U) Computing Resources

UNCLASSIFIED	
Resource	Description
AFSIM 2.2505	Licensed simulation framework (Linux or Windows)
C++ Compiler	GCC 11+ (Linux) or MSVC 2022 (Windows) with C++17
CMake 3.16+	Build system for plugin compilation
Python 3.12	Post-processing scripts
Workstation	Standard analyst workstation; no HPC required
	UNCLASSIFIED

8.3 (U) Documentation Deliverables

Table 17: (U) Documentation Deliverables

UNCLASSIFIED		
Document	Standard Reference	Status
V&V Plan (this document)	MIL-STD-3022, Appendix B	This document
V&V Report	MIL-STD-3022, Appendix C	To be produced
UAM	MIL-STD-3022, Appendix A	Complete (v3.0)
ADD	MIL-STD-3022	Complete (v2.0)
ICD	MIL-STD-3022	Complete (v1.0)
Accreditation Report	DoDI 5000.61	To be produced
		UNCLASSIFIED

8.4 (U) Level of Effort Estimate

Table 18: (U) Level of Effort

UNCLASSIFIED	
Activity	Estimated Effort
Code review (26 files, ∼7,736 LOC)	[PLACEHOLDER]
Unit test development and execution	[PLACEHOLDER]
Integration test development and execution	[PLACEHOLDER]
Validation test execution (16 tests)	[PLACEHOLDER]
V&V Report preparation	[PLACEHOLDER]
SME review and coordination	[PLACEHOLDER]
Total estimated effort	**[PLACEHOLDER]**
	UNCLASSIFIED

A (U) M&S Description

(U) The T-150 UAS AFSIM Plugin Suite v2.0 models the T-150/TRV-150 heavy-lift coaxial octocopter UAS within the AFSIM 2.2505 simulation framework. For complete model description, refer to the UAM v3.0, ADD v2.0, and ICD v1.0.

B (U) M&S Requirements Traceability Matrix

Table 19: (U) Requirements Traceability Matrix

UNCLASSIFIED				
Req ID	Requirement	Source	Test ID(s)	V&V Report Section
R-001	Battery SOC accuracy	First-principles	TF02-01	[TBD]
R-002	Voltage sag matches Li-Ion	Literature [9]	TF02-02	[TBD]
R-003	Motor thermal equilibrium	Thermodynamics	TF02-03	[TBD]
R-004	Position hold stability	Flight dynamics	TF01-01	[TBD]
R-005	Velocity tracking	Flight dynamics	TF01-02	[TBD]
R-006	Numerical stability	Software quality	TF01-05, TF03-03, TF04-03	[TBD]
R-007	VRS detection at thresholds	Aerodynamic theory	TF03-01, TF03-02	[TBD]
R-008	Ground effect below 1D	Cheeseman-Bennett [8]	TF01-04	[TBD]
R-009	Endurance within range	Manufacturer [11]	TF02-04	[TBD]
R-010	Slung load frequency	Classical mechanics	TF04-01, TF04-02	[TBD]
R-011	Attitude recovery	Flight dynamics	TF01-03	[TBD]
R-012	Cruise speed matches 30 m/s	Manufacturer [11]	TF01-02	[TBD]
R-013	Max payload 68 kg	Manufacturer [11]	TF02-05	[TBD]
				UNCLASSIFIED

C (U) Basis of Comparison

C.1 (U) Manufacturer Specifications

Table 20: (U) Manufacturer Specifications

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Parameter	Value	Source	V&V Usage
Cruise speed	30 m/s	[11]	R-012 validation
Max payload	68 kg	[11]	R-013 validation
MTOW	123 kg	[4]	Thrust/power calculations
Battery capacity	∼7 kWh	[5]	R-001, R-009 validation
Flight time	36 min	[11]	R-009 comparison
Motors	8x T-Motor U15II KV100	[4, 6]	Propulsion parameters
			UNCLASSIFIED

C.2 (U) Analytical Predictions

Table 21: (U) Analytical Predictions for Validation

UNCLASSIFIED			
Prediction	Formula	Parameters	Expected
Hover induced velocity	$v_i = \sqrt{mg/(2\rho A_{\text{disk}})}$	m=123 kg, ρ=1.225	∼8 m/s
Hover power	$P = mg \cdot v_i \cdot k$	k=1.15	17–20 kW
Pendulum frequency	$f = \frac{1}{2\pi}\sqrt{g/L}$	L=5 m	0.223 Hz
Ground effect (h=$R/2$)	$GE = 1/(1-(R/4h)^2)$	R=0.38	∼1.33
VRS onset	v_{desc} approaching v_i	threshold=4 m/s	50% of v_i
			UNCLASSIFIED

C.3 (U) Published Literature

Table 22: (U) Literature References

UNCLASSIFIED	
Reference	Application in V&V
Cheeseman & Bennett (1955) [8]	Ground effect thrust augmentation model
Johnson (1980) [9]	Momentum theory, VRS boundaries
Leishman (2006) [10]	VRS characterization, rotor aerodynamics
Li-Ion discharge curves (general)	OCV polynomial shape, resistance SOC dependence
	UNCLASSIFIED

D (U) References

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(U) [3] Department of the Army, "Army Directive 2024-02: Software Development," 2024.

(U) [4] Air Accidents Investigation Branch, "AAIB Investigation to Malloy Aeronautics T-150, South Scarle, Nottinghamshire," 2023.

(U) [5] Malloy Aeronautics, "Battery Pack Topology for Unmanned Aerial Vehicle," US Patent US20240262499A1, 2024.

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(U) [8] I.C. Cheeseman and W.E. Bennett, "The Effect of Ground on a Helicopter Rotor in Forward Flight," ARC R&M 3021, 1955.

(U) [9] W. Johnson, "Helicopter Theory," Princeton University Press, 1980.

(U) [10] J.G. Leishman, "Principles of Helicopter Aerodynamics," 2nd ed., Cambridge University Press, 2006.

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E (U) Acronyms

Table 23: (U) Acronyms

UNCLASSIFIED	
Acronym	Definition
ADD	Architecture Description Document
AFSIM	Advanced Framework for Simulation, Integration, and Modeling
AGL	Above Ground Level
AuxData	Auxiliary Data (AFSIM inter-processor communication)
C-UAS	Counter-Unmanned Aerial System
CG	Center of Gravity
CONOPS	Concept of Operations
DLL	Dynamic Link Library
DoD	Department of Defense
DoDI	DoD Instruction
ESC	Electronic Speed Controller
GE	Ground Effect
ICD	Interface Control Document
IGE	In Ground Effect
ISA	International Standard Atmosphere
KV	Motor velocity constant (RPM per Volt)
LOC	Lines of Code
M&S	Modeling and Simulation
MTOW	Maximum Takeoff Weight
NED	North-East-Down (coordinate frame)
OCV	Open-Circuit Voltage
OGE	Out of Ground Effect
PD	Proportional-Derivative (controller)
PID	Proportional-Integral-Derivative (controller)
RPM	Revolutions Per Minute
SME	Subject Matter Expert
SOC	State of Charge
TDD	Test Design Document
UAM	User and Analyst Manual
	UNCLASSIFIED

Continued on next page

Table 23: (U) Acronyms (Continued)

UNCLASSIFIED	
Acronym	Definition
UAS	Unmanned Aerial System
V&V	Verification and Validation
VRS	Vortex Ring State
VV&A	Verification, Validation, and Accreditation
6DOF	Six Degrees of Freedom
	UNCLASSIFIED

F (U) Acronyms

Citations